



CONTACT DETAILS

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SUMMARY

I work in representation theory and homological algebra, with an emphasis on categorical and cohomological methods for algebraic structures arising from symmetries. I obtained my PhD (August 2024) at the University of São Paulo under the supervision of Mikhailo Dokuchaev, where I developed a comprehensive homological framework for partial actions of groups and Hopf algebras, including structural and cohomological aspects of the associated algebraic constructions.

I am currently a Teaching Assistant at the Guangdong Technion–Israel Institute of Technology (GTIIT), where I conduct research under the supervision of María Julia Redondo. My research focuses on associative and graded algebras, with particular attention to deformation theory, derived invariants, and their interaction with representation-theoretic structures. I am interested in further developing my knowledge in areas such as homological algebra, algebraic geometry, and category theory and noncommutative geometry.

LANGUAGES

- Spanish (native)
- Portuguese (C1)
- English (C1)

EDUCATION

- **DOCTOR IN SCIENCE.** *University of São Paulo, Brasil.*
August 2020–August 2024
 - ◊ Thesis title: *Partial actions and homology.*
 - ◊ Scholarships: **CAPES** (Apr 2021 – Dec 2022) and **FAPESP** (Jan 2023 – Aug 2024)
 - ◊ Advisor: Mikhailo Dokuchaev
 - ◊ Achieved maximum grades in all courses
- **MASTER IN SCIENCE.** *University of São Paulo, Brasil.*
August 2018– August 2020
 - ◊ Dissertation title: *Partial group cohomology.*
 - ◊ Advisor: Mikhailo Dokuchaev
 - ◊ Scholarship: **CNPq** (Apr 2019 – Aug 2020)
 - ◊ Achieved maximum grades in all courses
- **BACHELOR’S DEGREE IN MATHEMATICS.** *Industrial University of Santander, Colombia.*
August 2014– July 2018
 - ◊ Dissertation title: *The inverse Möbius monoid*
 - ◊ Advisor: Hector Pinedo Tapia

PROGRAMMING SKILLS

- Python
- Rust
- C++

PUBLICATIONS

- M. DOKUCHAEV, E. JEREZ, *(Co)homology of partial smash products*, Journal of Algebra 652 (2024), 113–157.
- M. DOKUCHAEV, E. JEREZ, *The twisted partial group algebra and (co)homology of partial crossed products*, Bulletin of the Brazilian Mathematical Society 55 (2024), Article 33.
- E. JEREZ, *The category of partial group actions: quotients, (co)limits and groupoids*, Communications in Algebra (2025), 2528–2554.

PREPRINTS AND SUBMITTED WORKS

- E. JEREZ, *On the homology of partial group representations*, 2024.
- M. DOKUCHAEV, E. JEREZ, *Twisted partial group algebra and related topological partial dynamical systems*, 2024.
- E. JEREZ, *Homology of epsilon-strongly graded algebras*, 2025.
- M. DOKUCHAEV, E. JEREZ, J. L. VILCA-RODRÍGUEZ, *Globalization of partial group actions on non-associative algebras and covariant representations*, 2025.
- S. CAENEPEEL, M. DOKUCHAEV, E. JEREZ, H. PINEDO, *Partial group cohomology and the Amitsur complex*, 2026.
- M. DOKUCHAEV, E. JEREZ, E. N. MARCOS, *Globalization of Hochschild homology and smooth algebras*, 2026.
- E. JEREZ, M. J. REDONDO, L. ROMÁN, *Hochschild cohomology for infinitesimal deformations*, in preparation.
- M. DOKUCHAEV, E. JEREZ, H. PINEDO, *Partial actions of Hopf algebras: cohomology and Galois extensions*, in preparation.

EVENTS

TALKS:

- Partial group actions and groupoids.
XXVII BRAZILIAN ALGEBRA MEETING, SÃO PAULO, BRAZIL.
July 15 to 19, 2024.
- The partial group homology.
XXIV COLOQUIO LATINOAMERICANO DE ÁLGEBRA, SANTIAGO DE CHILE, CHILE.
July 22 to 26, 2024.
- **Invited talk** — Homology of epsilon strongly graded algebras.
CONFERENCE ON THEORETICAL AND COMPUTATIONAL ALGEBRA 2026, UNIVERSITY OF MINHO, GUIMARÃES, PORTUGAL.
June 28 to July 2, 2026.

POSTERS AND OTHERS:

- A homology theory for partial group actions. (Poster)
GROUPS, RINGS AND GROUP RINGS, UBATUBA, BRAZIL.
July 8 to 13, 2024
- 33^o COLÓQUIO BRASILEIRO DE MATEMÁTICA, IMPA, RIO DE JANEIRO.
August 2 to 6, 2021
- CIMPA SCHOOL: GEOMETRIC, ALGEBRAIC AND TOPOLOGICAL METHODS IN QUANTUM FIELD THEORY, VILLA DE LEYVA, COLOMBIA.
July 31 to August 09, 2023
- CIMPA SCHOOL: NON-ASSOCIATIVE ALGEBRAS AND RELATED TOPICS, VITORIA, BRAZIL.
December 4 to 9, 2023

SELECTED SEMINAR TALKS

- Twisted partial group algebras and homology of crossed products by twisted partial actions.
NONCOMMUTATIVE RINGS AND APPLICATIONS SEMINAR, UNIVERSITY OF SÃO PAULO, BRAZIL.
May 23, 2023.
- A homologia parcial de grupo.
REPRESENTATION THEORY OF ALGEBRAS AND APPLICATIONS SEMINAR, UNIVERSITY OF SÃO PAULO, BRAZIL.
August 16, 2024.
- Group cohomology and partial group cohomology.
NONCOMMUTATIVE RINGS AND APPLICATIONS SEMINAR, UNIVERSITY OF SÃO PAULO, BRAZIL.
September 24, 2024.
- Partial group actions and a homological globalization problem.
MCS COLLOQUIUM, GUANGDONG TECHNION-ISRAEL INSTITUTE OF TECHNOLOGY, CHINA.
December 18, 2025.
- Partial group actions and groupoids.
ALGEBRA SEMINAR, INDUSTRIAL UNIVERSITY OF SANTANDER, BUCARAMANGA, COLOMBIA.
February 18, 2026.

POSITIONS

- Postdoctoral Researcher
GUANGDONG TECHNION-ISRAEL INSTITUTE OF TECHNOLOGY (GTIIT), SHANTOU, CHINA.
Supervisor: María Julia Redondo.
2025 – 2026.
- Teaching Assistant / Tutor (concurrent position).
GUANGDONG TECHNION-ISRAEL INSTITUTE OF TECHNOLOGY (GTIIT), SHANTOU, CHINA.
2025 – 2026.

TEACHING EXPERIENCE

UNIVERSITY OF SÃO PAULO, BRAZIL

- **2023 - First Semester**
 - Course: MAT2110-2 - Calculus I for Chemistry
 - Period: February 1, 2023 - June 30, 2023, 6h/week.
- **2023 - Second Semester**
 - Course: MAT1352-3 - Calculus for Functions of a Real Variable II
 - Period: July 1, 2023 - November 30, 2023, 6h/week.
- **2024 - First Semester**
 - Course: MAT2453-3 - Differential and Integral Calculus I
 - Period: February 1, 2024 - June 30, 2024, 6h/week.

GUANGDONG TECHNION-ISRAEL INSTITUTE OF TECHNOLOGY

- **2025 - Spring Semester**
 - Course: 104016 - Algebra 1/Extended (Workshop - 4h/week).
 - Course: 104041 - Differential and Integral Calculus 1M (Workshop - 3h/week).
 - Course: 104043 - Differential and Integral Calculus 2M (Workshop - 5h/week).
- **2025 - Summer Semester**
 - Course: Pre Academic Math Course for Engineering Departments (Tutorial - 12h/week).
- **2025 - Winter Semester**
 - Course: 104281 - Infinitesimal Calculus 2 (Tutorial - 4h/week).
 - Course: 104066 - Algebra A (Tutorial - 6h/week and Workshop - 2h/week).
- **2026 - Spring Semester**
 - Course: 104066 - Algebra A (Tutorial - 6h/week and Workshop - 2h/week).
 - Course: 104142 - Metric and topological spaces (Tutorial - 1h/week).